FIM549 Mini Project

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March 12, 2026

Data Preparation

Historical market data was sourced using the `yfinance` library in Python, which pulls from Yahoo Finance's API. The dataset consists of daily adjusted closing prices for any stock and ETF needed.

The data spans from **January 3rd, 2023, to February 23rd, 2026** for INTC and SPY. To prepare the data for modeling, daily prices were converted into logarithmic returns.

Calculation Methods

The following calculations were used in the report:

- **Return Calculation:** Daily returns were calculated using logarithmic change $\ln(P_t/P_{t-1})$. This ensures that the returns are symmetric, which is a requirement for the assumptions underlying the GARCH models and the normal distribution. Simple returns were used for the average daily return and average annual return metrics.

- **Volatility Estimation:** Since markets experience volatility clustering, an average is insufficient for forecasting risk. The Generalized Autoregressive Conditional Heteroskedasticity (GARCH 1,1) model was fitted to the return data. This model estimates current volatility by weighting long-term average variance, recent market shocks, and past forecasted variance.

- **Historical Value at Risk, Expected Shortfall:** Historical VaR was calculated using the Historical Simulation approach. The simulated daily portfolio returns over the entire timeframe were sorted from worst to best. The 99% VaR was found using the cutoff point for the worst 1% of historical days. Expected Shortfall was then calculated by taking the average of all daily losses that exceeded this VaR threshold, to measure tail risk.

- **Normal VaR:** Normal VaR was estimated assuming a normal distribution. The combined portfolio variance was calculated using the GARCH-forecasted volatilities and the historical correlation. Then the portfolio standard deviation was multiplied by the Z-score for a 99% confidence interval.

- **Risk Adjusted Performance:** Return on Risk Adjusted Capital was calculated for each asset. This was done by dividing the expected daily return of the asset by its 99% Historical VaR, serving as the Economic Capital requirement.

Returns and GARCH Volatility Estimates

Log Returns and Average Returns

The daily returns for both assets were calculated using the historical closing prices. The total returns from January 3rd, 2023, to February 23rd, 2026, and the average daily returns are:

Metric	INTC	SPY
Total Return	71.13 %	88.33 %
Average Daily Return	0.0685 %	0.0807 %
Average Annual Return	18.82 %	22.53 %

Table 1: Historical Return Metrics

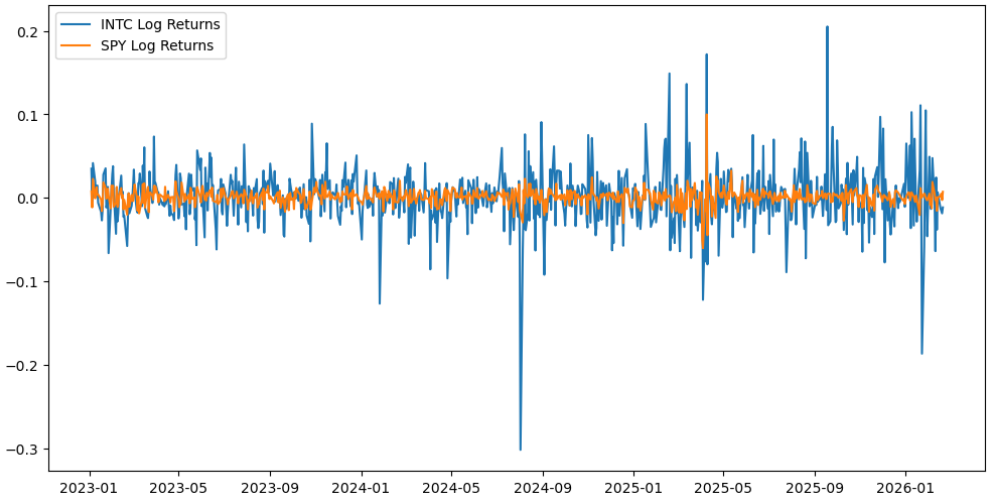

Figure 1: Daily Log Returns Over Time

Volatility Estimates as of February 23, 2026

Using a GARCH(1,1) model, the conditional volatility on the final day of the dataset, February 23rd, 2026, was estimated at:

- **INTC Volatility:** 0.028532
- **SPY Volatility:** 0.008363

GARCH(1,1) Model Parameters and Evaluation

Model Parameters, P-values

The GARCH(1,1) model is: $\sigma_t^2 = \omega + \alpha\mu_{t-1}^2 + \beta\sigma_{t-1}^2$ with $\omega = \gamma V_L$. The estimated parameters and their corresponding p-values are listed below.

	INTC		SPY	
Parameter	**Estimate**	**P-value**	**Estimate**	**P-value**
μ	-0.000022	0.985753	0.000945	0.000000
ω	0.000401	0.000173	0.000009	0.000000
α	0.380171	0.104439	0.100000	3.726911e-03
β	0.358789	0.012617	0.800000	3.088922e-264

Table 2: GARCH(1,1) Parameters and Significance

Models Evaluation: For INTC, the p-values for α and μ are both greater than 0.05, meaning they are not statistically significant at the 95% confidence level. For SPY, all of the parameters are less than 0.05, so they are statistically significant at the 95% confidence level.

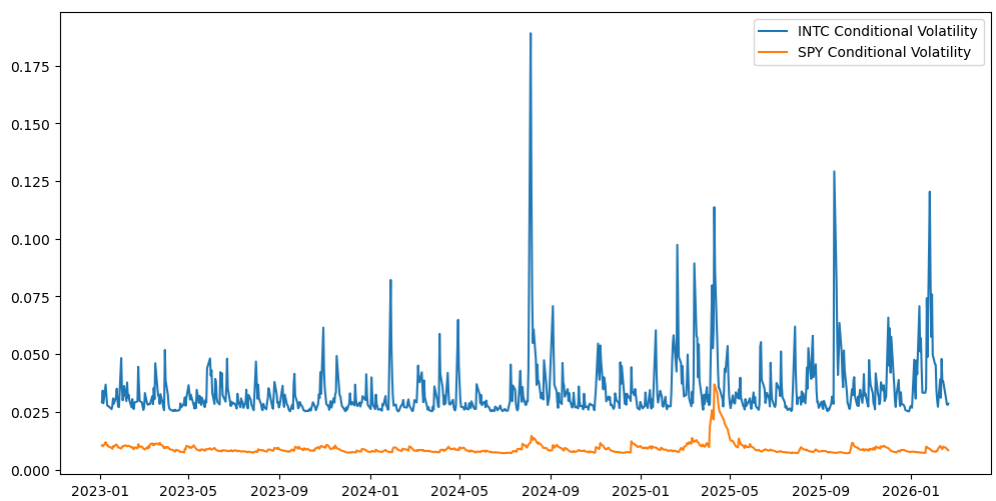

Figure 2: Conditional Volatility Over Time

Volatility Forecast

Generating a 3-day forecast from the GARCH(1,1) model gives the expected volatilities for the third day after February 23, 2026:

- **INTC Forecasted Volatility:** 0.033211
- **SPY Forecasted Volatility:** 0.008552

Correlation Estimate

The historical correlation ρ between the daily log returns of INTC and SPY over the analyzed period is **0.4562**.

Historical VaR, ES, and Diversification

The following risk metrics are calculated for three specific portfolios, assuming the exposure values for every day, using historical simulation at a 99% confidence level.

Portfolio	Exposure	1-Day 99% VaR	1-Day 99% ES
INTC Only	\$1,000,000	\$85,284.95	\$140,831.15
SPY Only	\$1,000,000	\$26,635.60	\$39,599.67
Combined	\$2,000,000	\$89,174.80	\$154,994.38

Table 3: Historical VaR and Expected Shortfall

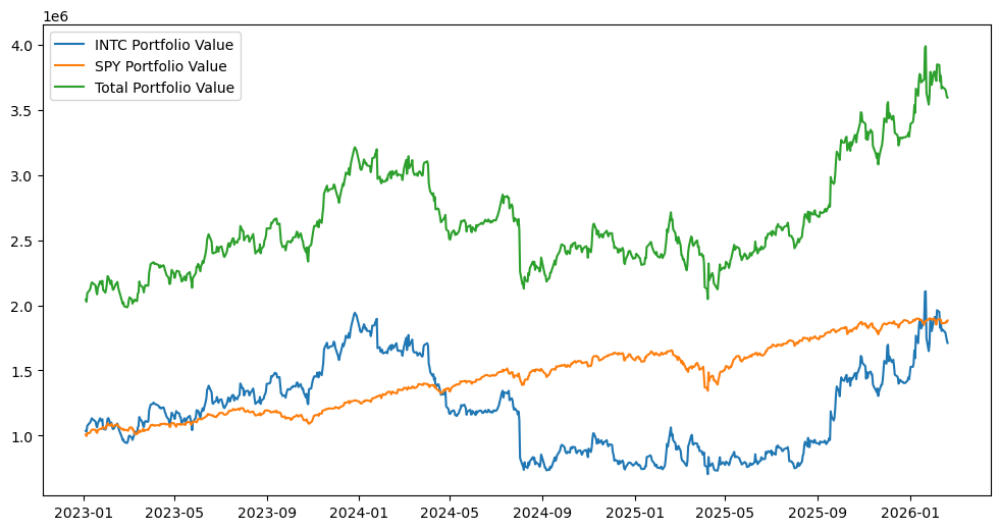

Figure 3: Simulated Portfolios Over Time

Diversification Benefit

The diversification benefit is the difference between the sum of the individual VaRs and the combined portfolio VaR:

$$\text{Benefit} = (VaR_A + VaR_B) - VaR_{A+B} = \$85,284.95 + \$26,635.60 - \$89,174.80$$

The total diversification benefit is **\$22,745.76**. This risk reduction occurs because the two assets are not perfectly correlated since $\rho < 1$.

Normal VaR Comparison

Normal Distribution 99% VaR

Using the daily volatilities forecasted by the GARCH model and the historical correlation, the dollar variance of the Total Portfolio was calculated. Applying the standard normal Z-score for a one-tailed 99% confidence interval $Z \approx 2.326$, the Normal VaR is:

- **1-Day 99% Normal VaR:** $74,246.10

Normal vs. Historical VaR

Comparing the two methods for the Total Portfolio:

- **Historical VaR:** $89,174.80
- **Normal VaR:** $74,246.10

The **Historical** approach yields a larger VaR estimate.

Financial returns exhibit excess kurtosis and negative skewness, meaning extreme market downturns happen more frequently than a normal distribution. The Historical Simulation captures these actual extreme past events, resulting in a larger VaR. The Normal distribution underestimates the probability of events in the tail, giving a lower required capital buffer.

Risk-Adjusted Performance

To assess which investment performs better per unit of risk, we calculate the Return on Risk-Adjusted Capital (RORAC), using the 99% Historical VaR as the economic capital requirement.

Portfolio	Expected Return ($)	Economic Capital ($)	Risk-Adjusted Ratio
INTC	$684.43	$85,284.95	0.008025
SPY	$806.42	$26,635.60	0.030276
Combined	$1,490.85	$89,174.80	0.016718

Table 4: Risk-Adjusted Performance Comparison

The portfolio of **SPY** has superior risk-adjusted performance. For every dollar of economic capital held to cover downside risk, it generates higher expected returns.